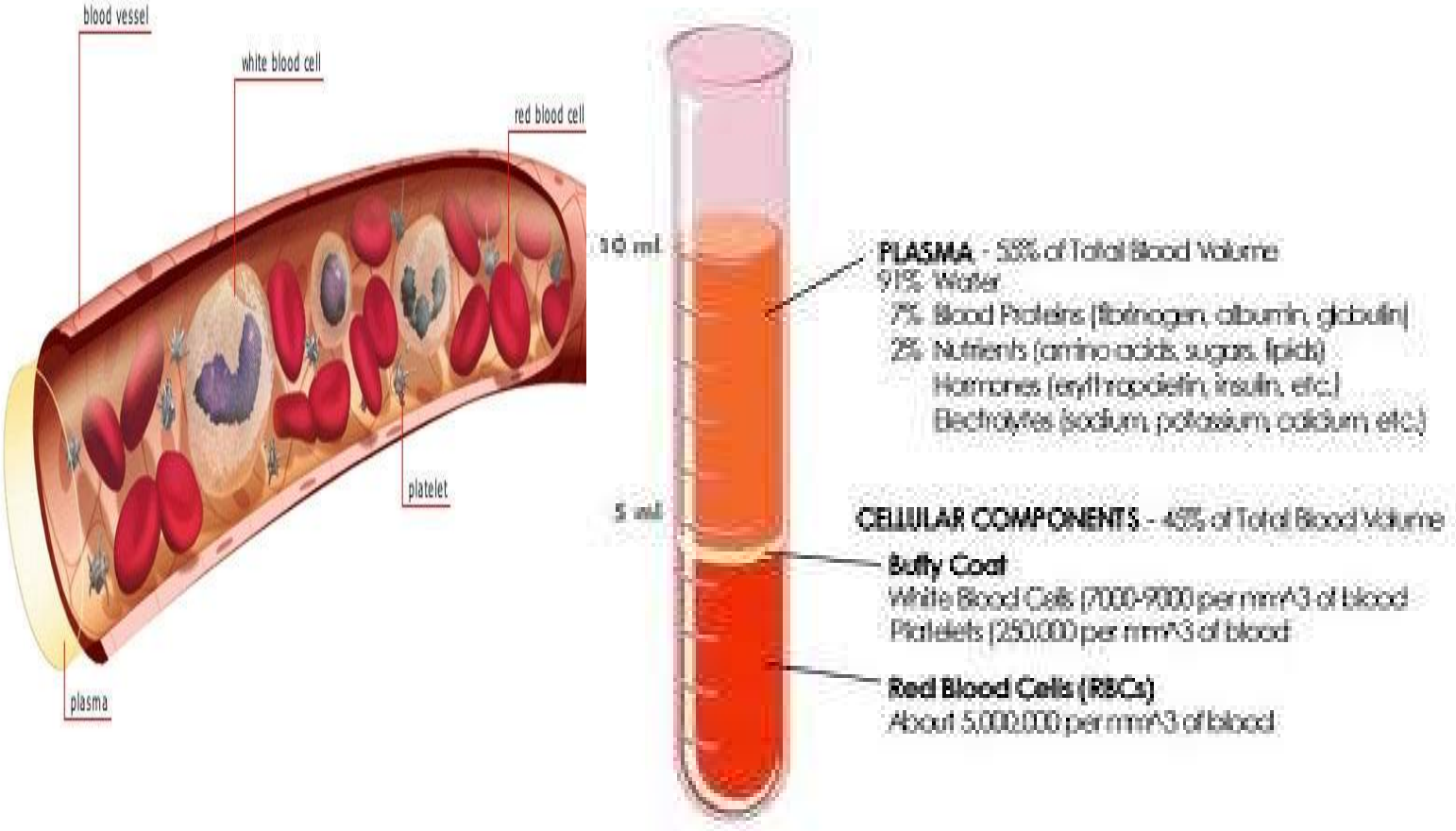
Plasma Proteins, Immunoglobulins and their Clinical and Biomedical Significance respectively



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Plasma proteins

- “Plasma proteins are proteins present in blood plasma”.
- Plasma contains a complex mixture of proteins.
- Early scientists classified these proteins into three groups, fibrinogen, albumin, and globulins, on the basis of their physicochemical properties.
- Subsequently, based on electrophoresis technique, salt-soluble serum protein fraction separated into five major components designated albumin and the α1 -, α2 -, β-, and γ globulins.
- Plasma proteins frequently contain bound carbohydrates (glycoproteins) or lipids (lipoproteins).

Plasma proteins: Roughly 70 to 80% of all plasma proteins are synthesized in the liver.

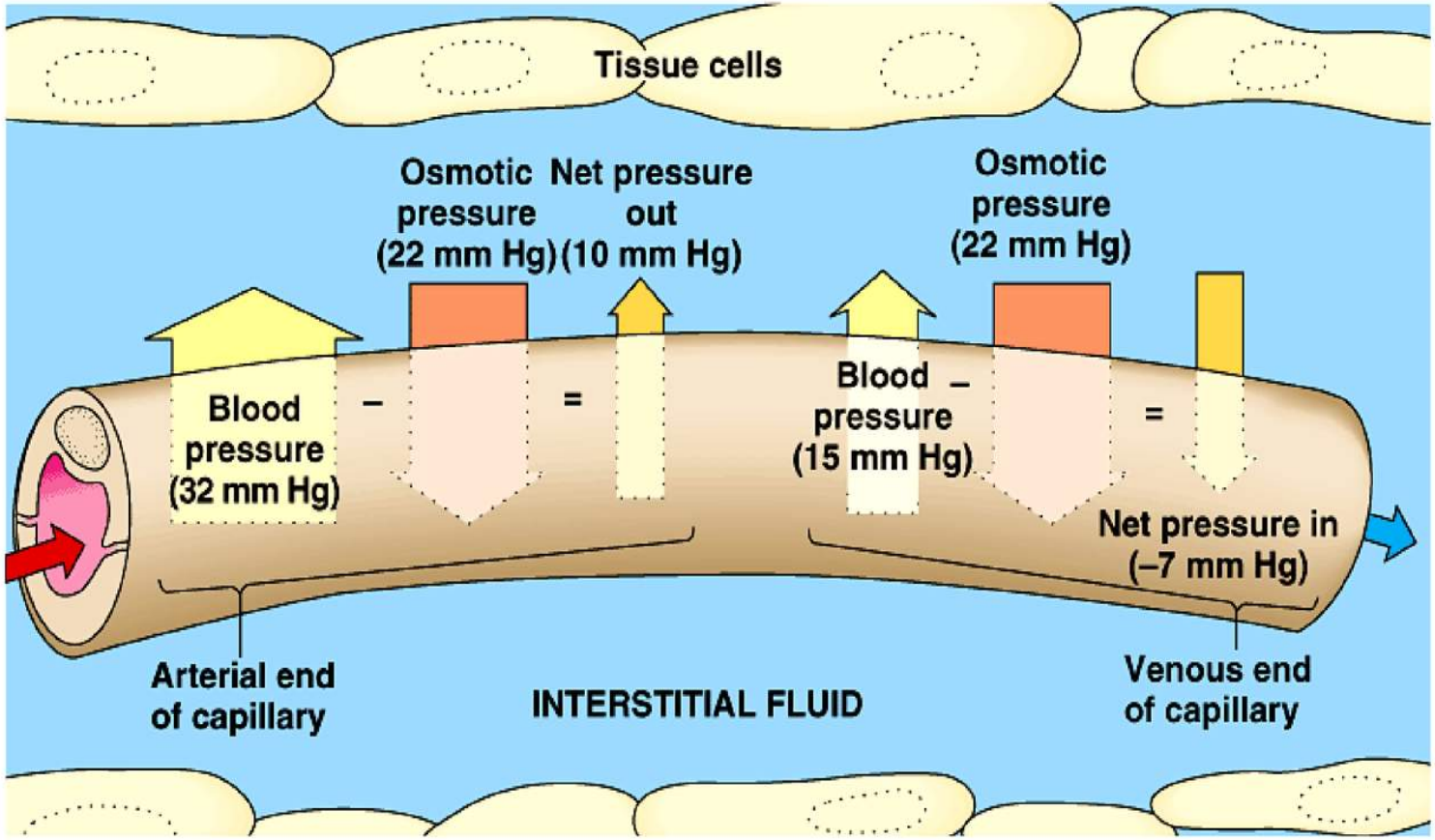
Plasma Proteins			
Protein	**Percentage of Total**	**Origin**	**Function**
Albumin	60%	Liver	Helps maintain blood osmotic pressure
Globulin	36%		
Alpha globulins		Liver	Transport lipids and fat-soluble vitamins
Beta globulins		Liver	Transport lipids and fat-soluble vitamins
Gamma globulins		Lymphatic tissues	Constitute a type of antibody
Fibrinogen	4%	Liver	Blood coagulation

- The aggregate concentration of the proteins present in human plasma typically falls in the range of 7 to 7.5 g/dL (normal laboratory value of plasma total protein).

Blood protein	Normal level
Albumins	**3.4-4.7 g/dL**
Globulins	**2.0-2.5 g/dl**
Fibrinogen	**0.2-0.45 g/dl**
The Albumin : Globulin ratio is normally	2:1

Plasma proteins carry out a number of different functions.

- Plasma proteins help determine the distribution of fluid between blood & tissues



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Edema

- If the concentration of plasma proteins is markedly diminished, fluid will cease flowing back into the intravascular compartment and begin to accumulate in the extravascular tissue spaces, resulting in a condition known as edema.



ALBUMIN IS THE MOST ABUNDANT PROTEIN IN HUMAN PLASMA

- It accounts for roughly **three-fifths** of total plasma protein by weight (3.4-4.7 g/dL).
- Half-life: 20 days. For this reason, measurement of serum albumin concentration is used to assays liver function test.
- Shape: ellipsoid stabilized by a total of 17 intrachain disulfide bonds.
- Mature human albumin consists of a single polypeptide chain, 585 amino acids in length.

Functions

Transport:

- A major role of albumin is to bind and transport many diverse molecules which include:
  - a. *Metal ions:* such as calcium and copper.
  - b. *Free fatty acid:* albumin binds to free fatty acid released by adipose tissue and facilitates their transfer to other tissue.
  - c. *Bilirubin:* this protects from the toxic side effects of unconjugated bilirubin.
  - d. *Bile acid:* albumin carries the bile acids that are recycle from the intestine to the liver in the hepatic portal vein.
  - e. *Hormones:* such as thyroid hormones and the steroid hormones.
  - f. A variety of drugs, including sulfonamides, penicillin G, and aspirin, also bind to albumin.

Acute-phase Proteins

- The levels of certain plasma proteins increase during inflammation or following tissue damage.
- C-reactive protein (CRP, so named because it reacts with the C polysaccharide of pneumococci), α1 -antiproteinase, haptoglobin, α1 -acid glycoprotein, and fibrinogen are classified as acute-phase proteins.
- Acute-phase proteins are believed to play a role in the body's response to inflammation. C-reactive protein stimulates the complement pathway, while α1 -antitrypsin neutralizes certain proteases released during acute inflammation.

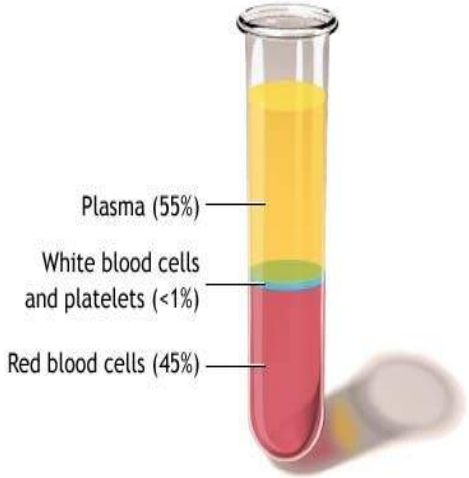
PLASMA PROTEINS	FUNCTION
Pre-albumin	Binds T3 & T4
Albumin	Transport , colloid oncotic pressure
α1-globulin : α1-antitrypsin	Antiproteinase
α2 - globulins Ceruloplasmin haptoglobulin Prothrombin α2-macroglobulin	 Copper transport Binds haemoglobin Haemostasis Transport anti--proteinase
β—Globulins Transferrin Plasminogen Fibrinogen	 Iron –transport Fibrinolysis Haemostasis

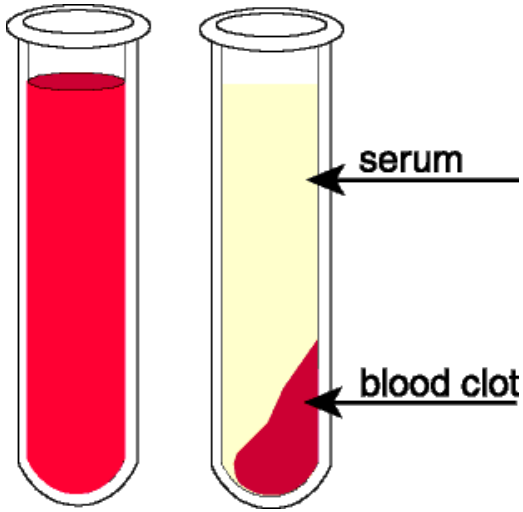
γ-Globulin **IgA** **IgM** **IgG** **IgE** **IgD**	**Immunity**
Clotting factors	**Conversion of fibrinogen into Fibrin**
Lipoproteins (chylomicrons, VLDL, LDL, HDL)	**Transport lipids**

Other Functions

- Buffer action
- Protein reserve
- Act as precursors of active substances. E:g angiotensinogen to angiotensin.
- Maintain viscosity

Plasma vs. serum

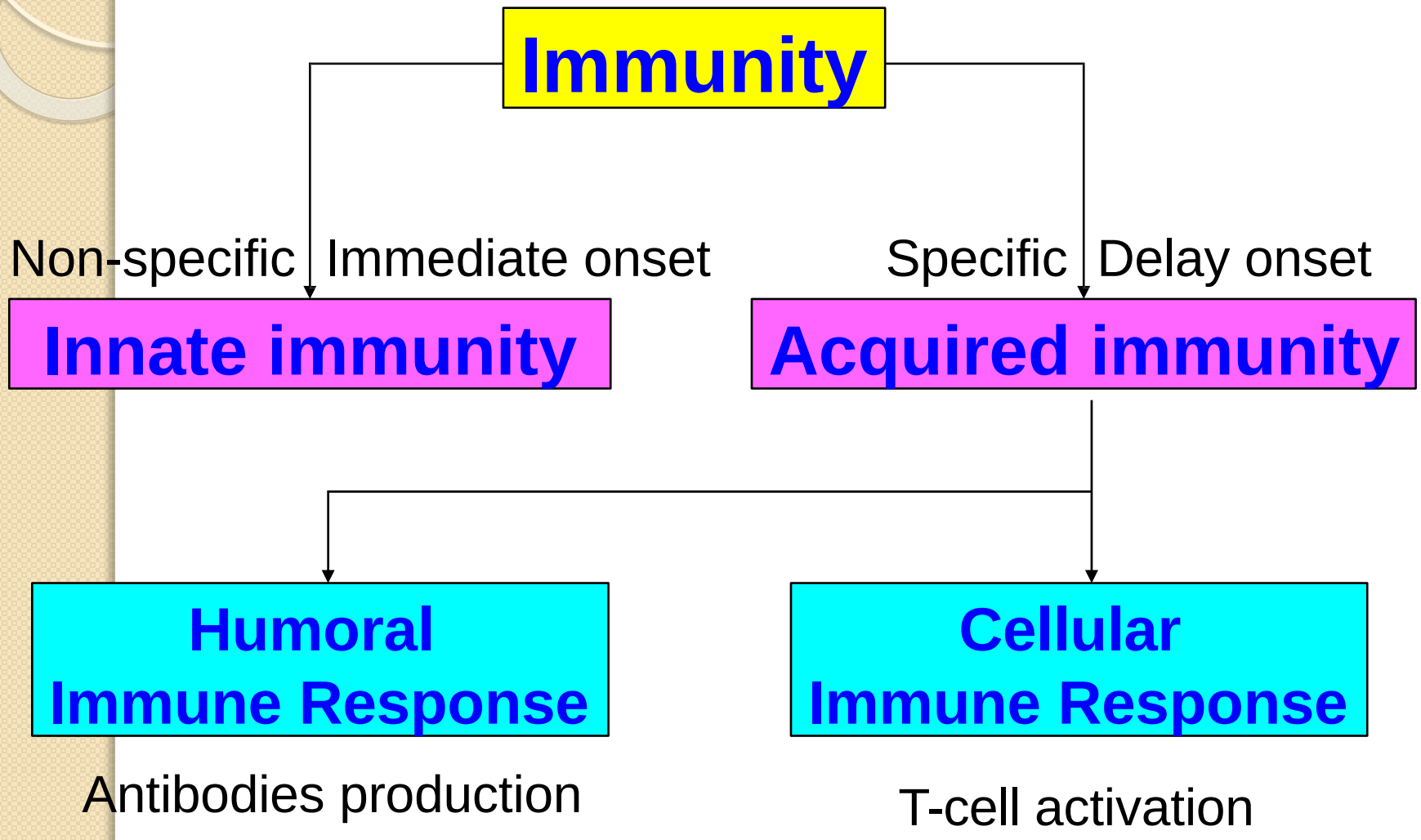




serum= plasma - fibrinogen

Immunoglobulins

What types of immune responses protect individuals from infections?



3 salient features of immunological reaction

- Recognition of self
- Specificity
- Memory of response

Antibodies (Ab)

- Proteins that recognize and bind to a particular antigen with very high specificity.

- Belong to a group of serum proteins called immunoglobulins (Ig)

HUMORAL IMMUNITY

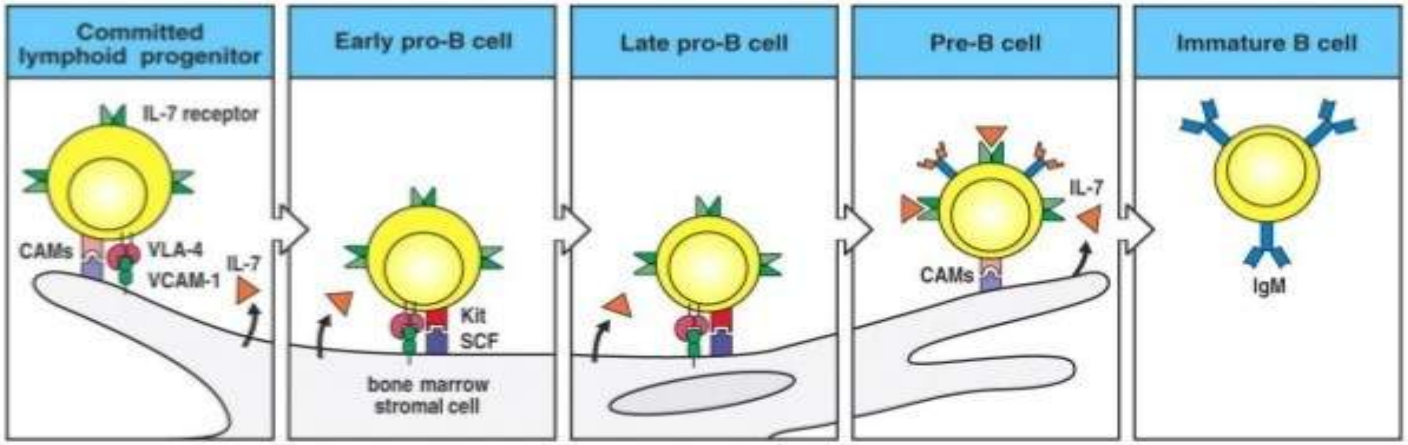
- Antibodies are produced after stimulation of B lymphocytes by antigens in peripheral lymphoid organs (lymph nodes, spleen, mucosal lymphoid tissues) and at tissue sites of inflammation.

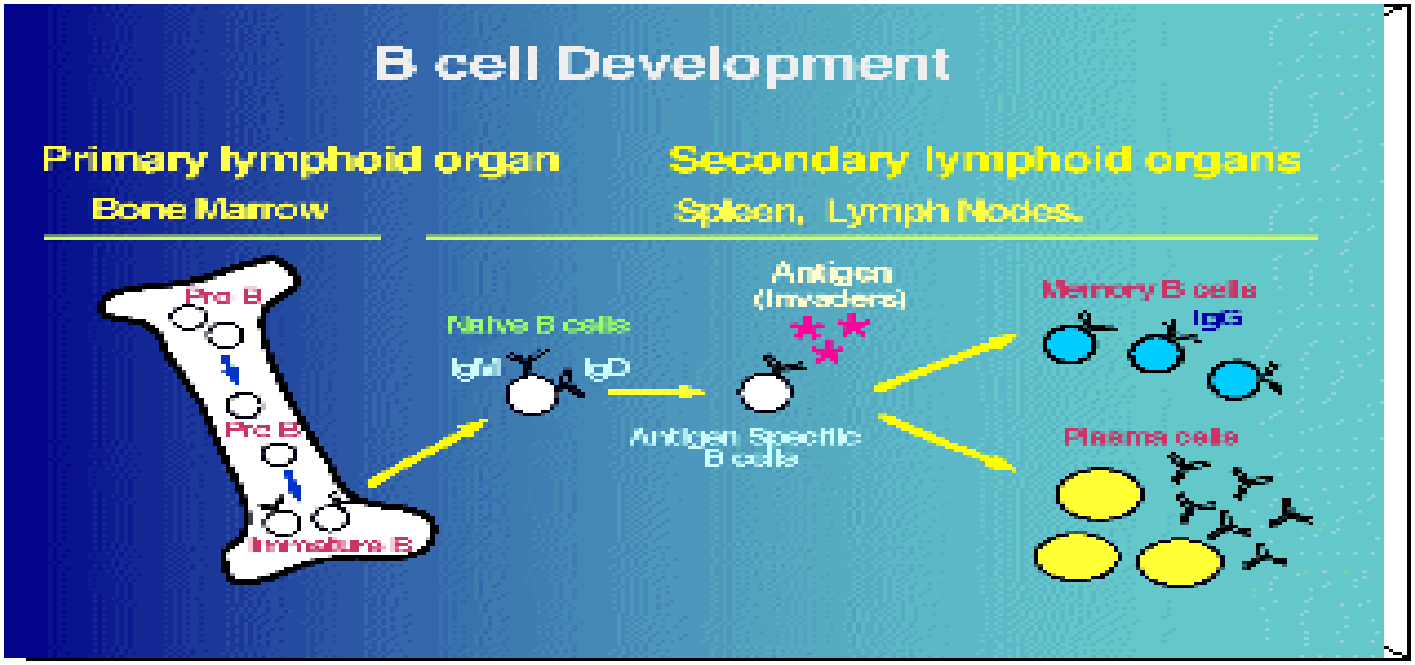
- Many of the antigen-stimulated B lymphocytes differentiate into antibody-secreting plasma cells, some of which remain in lymphoid organs or inflamed tissues and others migrate to and reside in the bone marrow.

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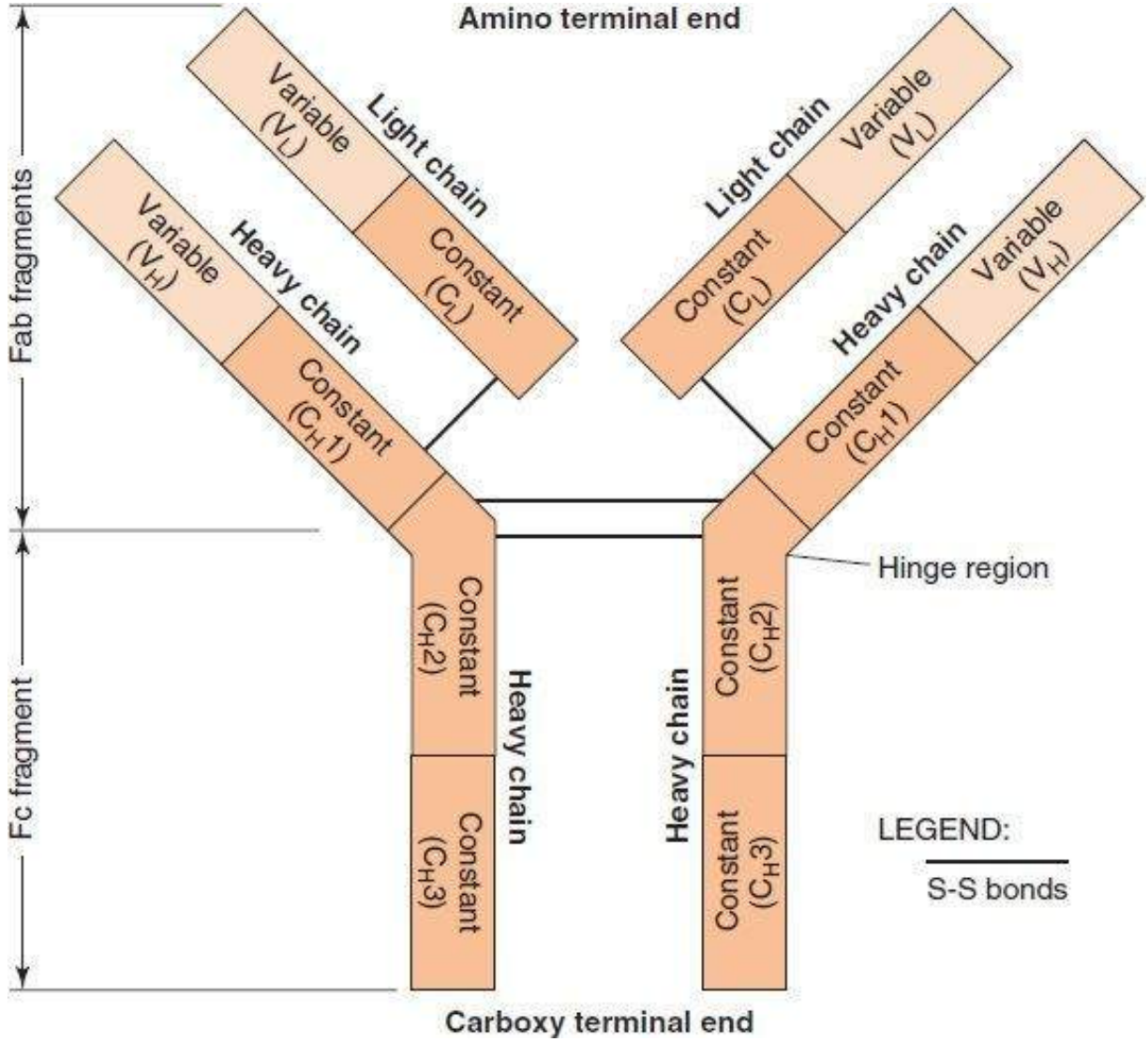
- Plasma cells synthesize and secrete antibodies of different heavy-chain isotypes (classes).
- These secreted antibodies enter the blood, from where they may reach any peripheral site of infection, and enter mucosal secretions, where they prevent infections by microbes that try to enter through epithelia.
- Thus, antibodies are able to perform their functions throughout the body.

B cell Development





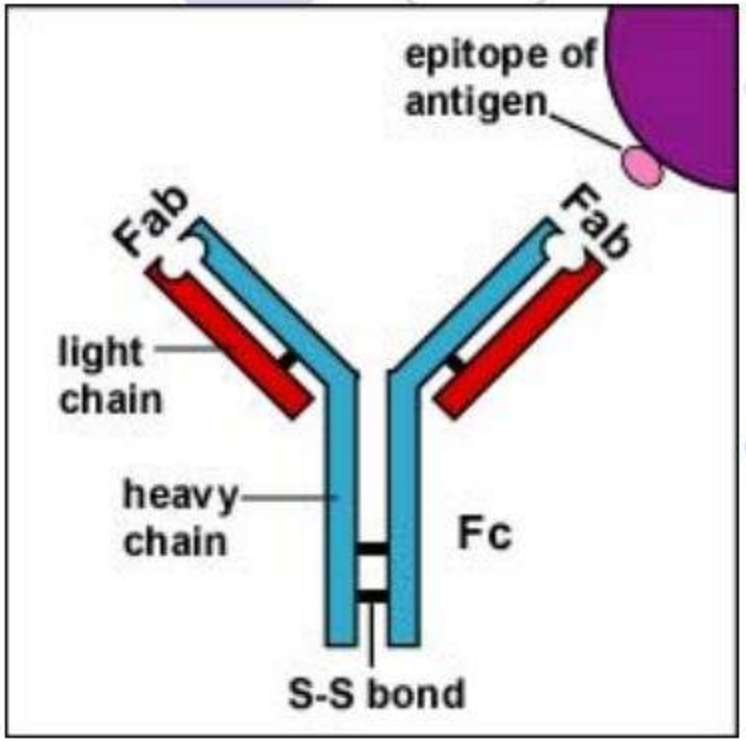
IMMUNOGLOBULIN STRUCTURE



Immunoglobulins

- IgG - Gamma heavy chains
- IgM - Mu heavy chains
- IgA - Alpha heavy chains
- IgD - Delta heavy chains
- IgE - Epsilon heavy chains
- Two types of light chains :
- kappa and lambda
- An Ig contains either two kappa or two lambda light chain and never a mixture.
- Kappa chain (60%) is more common in human.

- Antibodies use their antigen-binding (Fab) regions to bind to and block the harmful effects of microbes and toxins, and they use their Fc regions to activate diverse effector mechanisms that eliminate these microbes and toxins

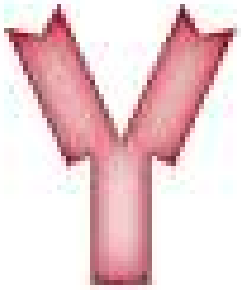


Properties of Human Ig

Property	IgG	IgA	IgM	IgD	IgE
Percentage of total immunoglobulin in serum (approx)	75	15	9	0.2	0.004
Serum concentration (mg/dL) (approx)	1000	200	120	3	0.05
Sedimentation coefficient	7S	7S or 11S[1]	19S	7S	8S
Molecular weight (×1000)	150	170 or 400[1]	900	180	190
Structure	Monomer	Monomer or dimer	Monomer or pentamer	Monomer	Monomer
H chain symbol	γ	α	μ	δ	ε
Complement fixation	+	–	+	–	–
Transplacental passage	+	–	–	–	–
Mediation of allergic responses	–	–	–	–	+
Found in secretions	–	+	–	–	–
Opsonization	+	–	–[2]	–	–
Antigen receptor on B cell	–	–	+	?	–
Polymeric form contains J chain	–	+	+	–	–

[1]The 11S form is found in secretions (e.g., saliva, milk, and tears) and fluids of the respiratory, intestinal, and genital tracts.

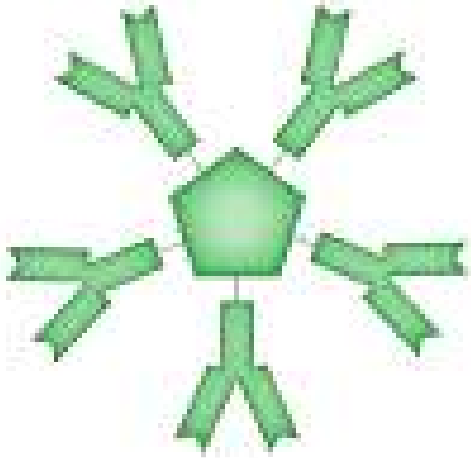
[2]IgM opsonizes indirectly by activating complement. This produces C3b, which is an opsonin.



Monomer
IgD, IgE, IgG



Dimer
IgA



Pentamer
IgM

Rh iso immunisation

- This occurs when mother is Rh negative and fetus is Rh positive.
- During parturition, fetal RBCs may enter into maternal circulation , leading to formation of anti-Rh antibodies.
- During next pregnancy, these Ab being IgG in class, can enter into fetal circulation causing fetal hemolysis, neonatal jaundice and in severe cases, neonatal death or miscarrige.
- Anti-Rh Ab injected within 24 hrs of delivery of first child , will avert the isoimmunization and protect future pregnancy

IgG

- Opsonization effect –
- IgG – Fab can bind with microbes, and also the Fc portion can fix with macrophage, so that phagocytosis is made easier
- Antibody dependent cell mediated cytolysis (ADCC) - Antibodies serve as bridge b/w the infected target cell and effector cell.
- ADCC effector cell (Natural killer cells) secrete lytic enzymes – lysis of infected target cell.

Important Functions of Immunoglobulins

Immunoglobulin	Major Functions
IgG	Main antibody in the secondary response. Opsonizes bacteria, making them easier to phagocytize. Fixes complement, which enhances bacterial killing. Neutralizes bacterial toxins and viruses. Crosses the placenta.
IgA	Secretory IgA prevents attachment of bacteria and viruses to mucous membranes. Does not fix complement.
IgM	Produced in the primary response to an antigen. Fixes complement. Does not cross the placenta. Antigen receptor on the surface of B cells.
IgD	Uncertain. Found on the surface of many B cells as well as in serum.
IgE	Mediates immediate hypersensitivity by causing release of mediators from mast cells and basophils upon exposure to antigen (allergen). Defends against worm infections by causing release of enzymes from eosinophils. Does not fix complement. Main host defense against helminth infections.

Plasma cell disorders:

Def: “Plasma cell disorders are neoplastic proliferations of well-differentiated immunoglobulin producing cells”.

- They are usually present within the bones as discrete masses and cause systemic symptoms related the production of a monoclonal immunoglobulin.
- They may or may not secrete detectable levels of a monoclonal immunoglobulin or immunoglobulin fragment (M protein).
- They are most common in middle-aged and elderly persons (40-50yrs).

Variants of PCD

- Six major variants:
  1. Multiple myeloma
  2. Localized plasmacytoma (solitary myeloma)
  3. Lymphoplasmacytic Lymphoma
  4. *Heavy-chain disease*
  5. *Monoclonal gammopathy of undetermined significance (MGUS)*
  6. *Primary or immunocyte-associated amyloidosis*

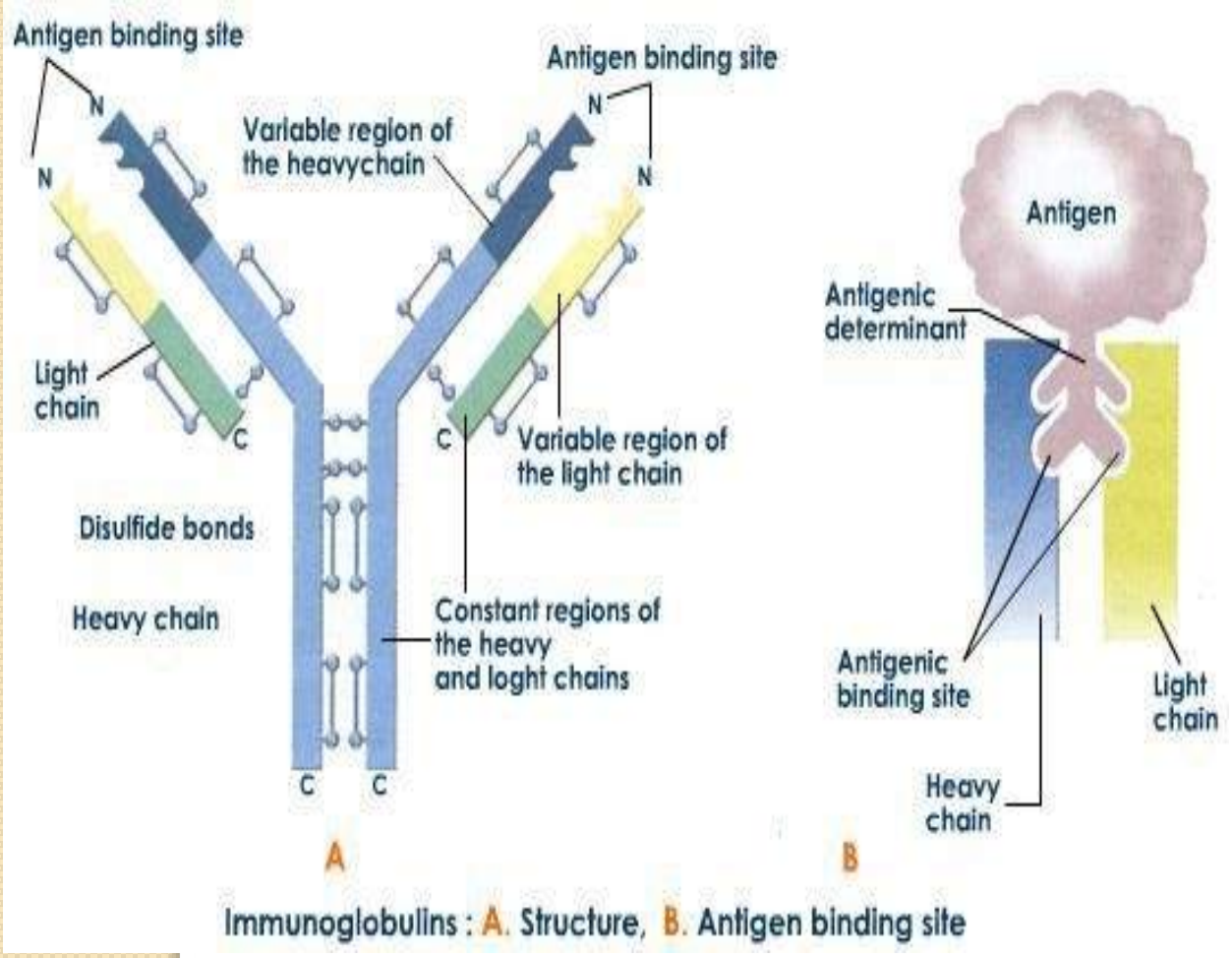
MULTIPLE MYELOMA

- Synonym: **Plasma cell myeloma/Plasmacytoma**
- By far the **most common** of the malignant PCD.
- The peak age of incidence of MM is b/w **50-60yrs**.

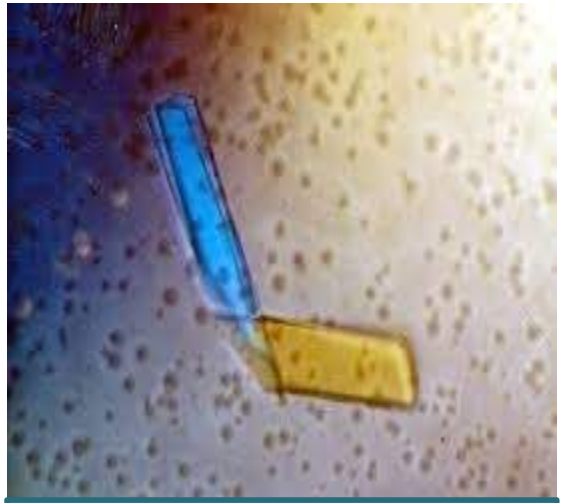
- Definition:
  - **"Malignant plasma cell tumor that typically involves bone and is associated with prominent serum and urinary protein abnormalities"**

Protein abnormalities

- The tumor cells produce massive quantities of identical Ig molecules demonstrable electrophoretically referred to as an **M protein/component**.
- The most common M component **(in serum)** is **IgG (60%),**
  - followed by IgA (20-25%);
  - only rarely is it IgM, IgD or IgE.
  - In the remaining 15-20%of cases, the plasma cell produce only light chains.
- Because of their low molecular weight, the free light chains are rapidly excreted in the **urine**, where they are termed as “BENCE-JONES PROTEINS”(Def: immunoglobulin light chain found in the urine)



Immunoglobulins : A. Structure, B. Antigen binding site



BENCE-JONES PROTEINS

- C - Calcium is increased (Hypercalcemia)
- R – Renal failure
- A – Anemia
- B – Lytic Bone lesion(bone pain & tenderness) - 70%
- Spontaneous pathological fracture of weight bearing bones, rib and vertebrae may occur.
- Paraproteinemia (monoclonal Ig) – leads to Proteinuria
- Recurrent infections are common - normal Ig is diminished
- Bone marrow examination – malignant plasma cells • X ray shows “punched-out osteolytic lesions”

Amyloidosis

- About 20 % patients with myeloma develop amyloidosis
- deposits of light chain fragments in tissue (liver, kidney, intestine) of MM pateints.


Thank You